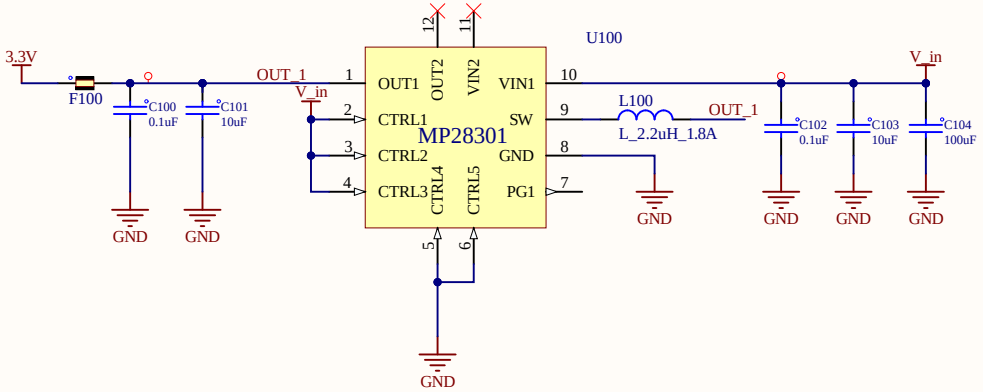
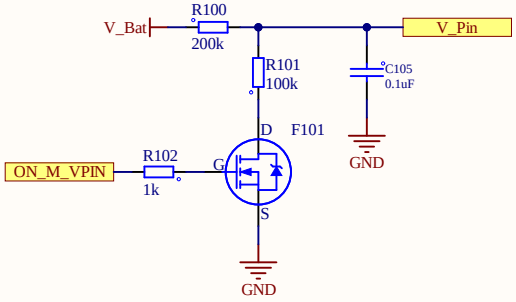
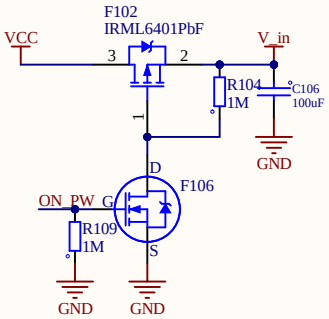
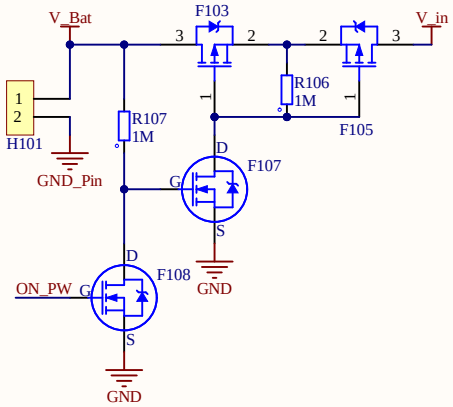


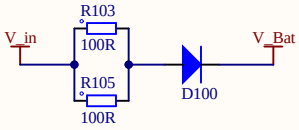


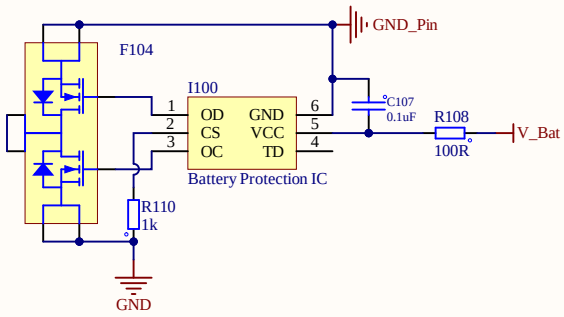




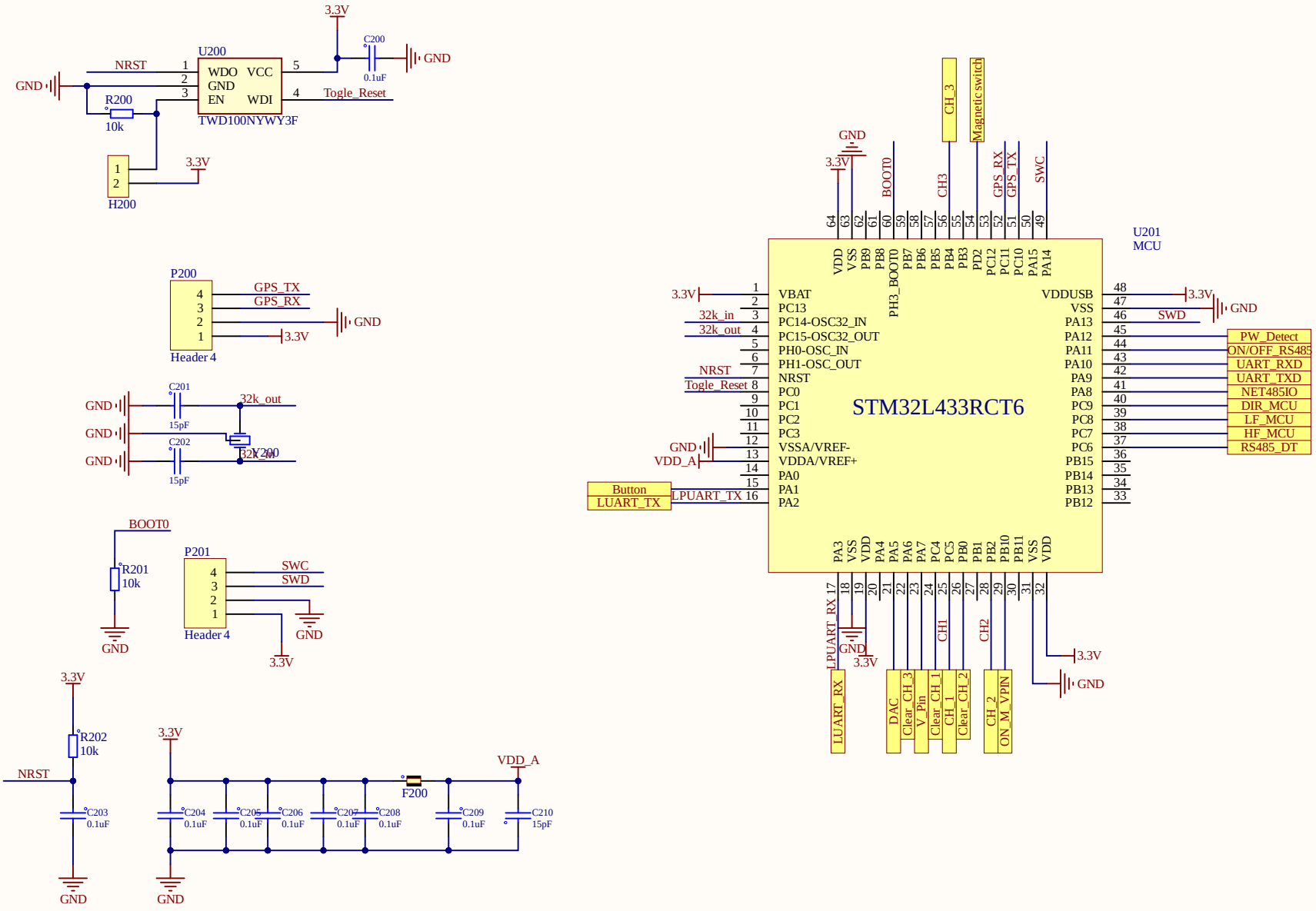


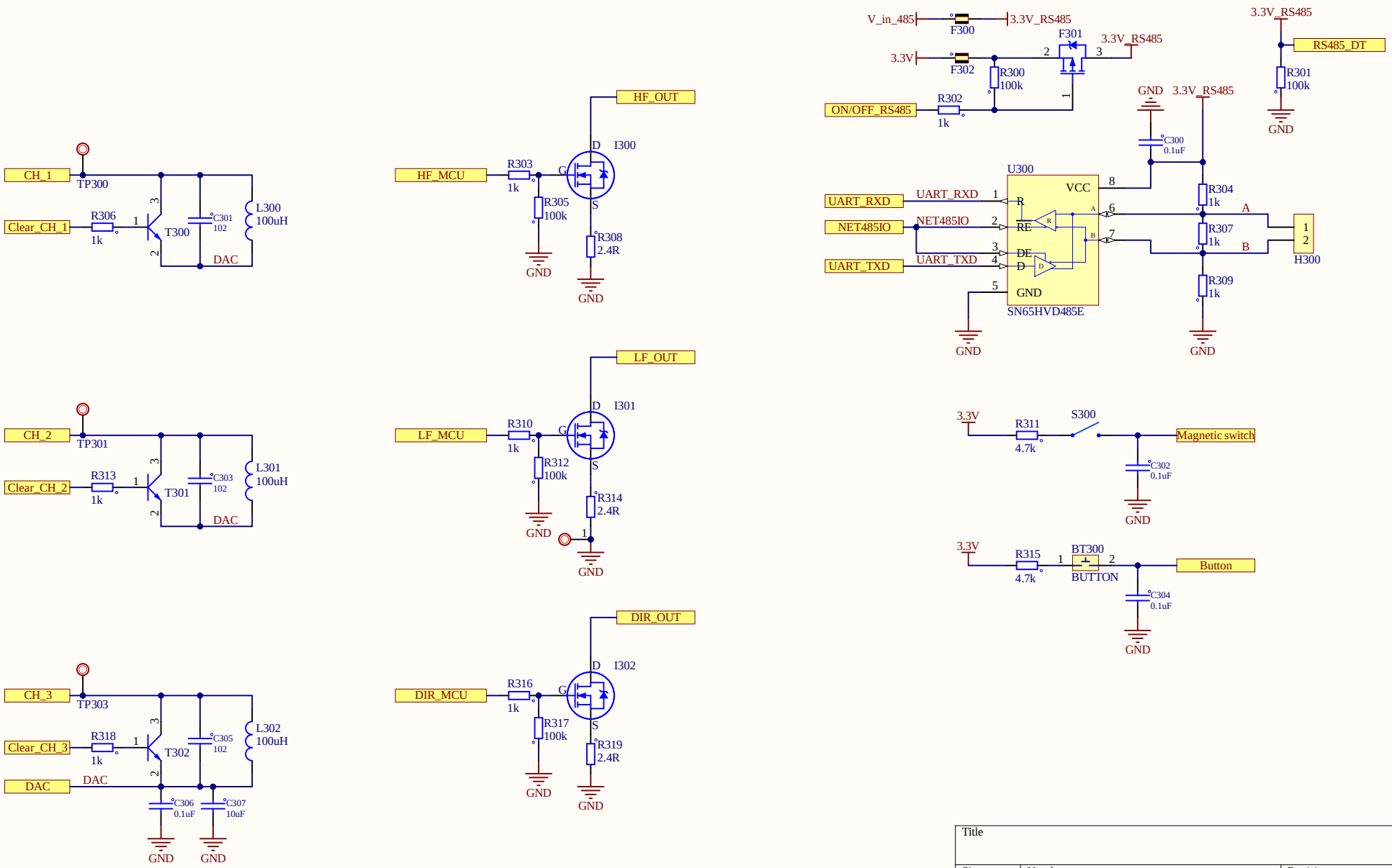


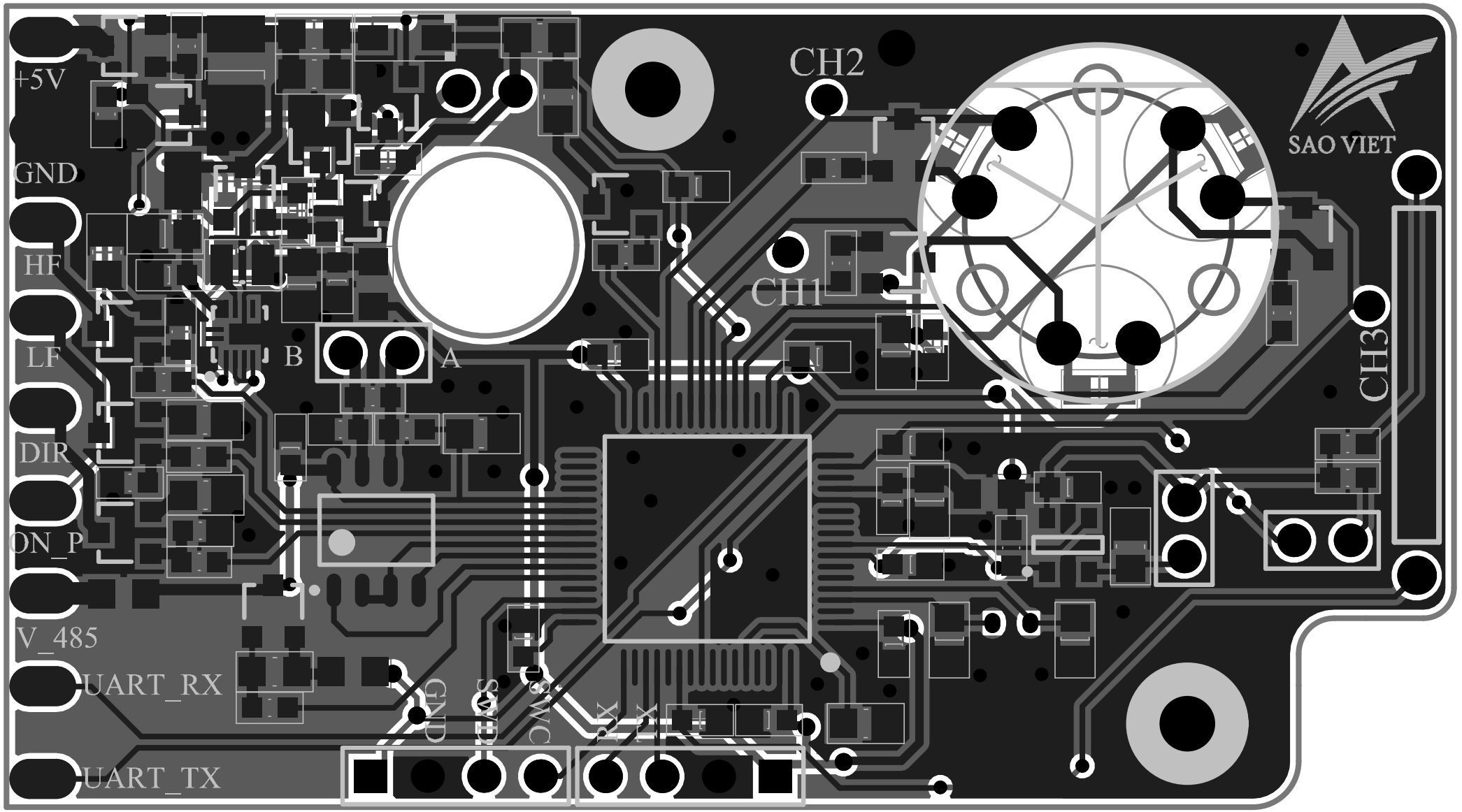


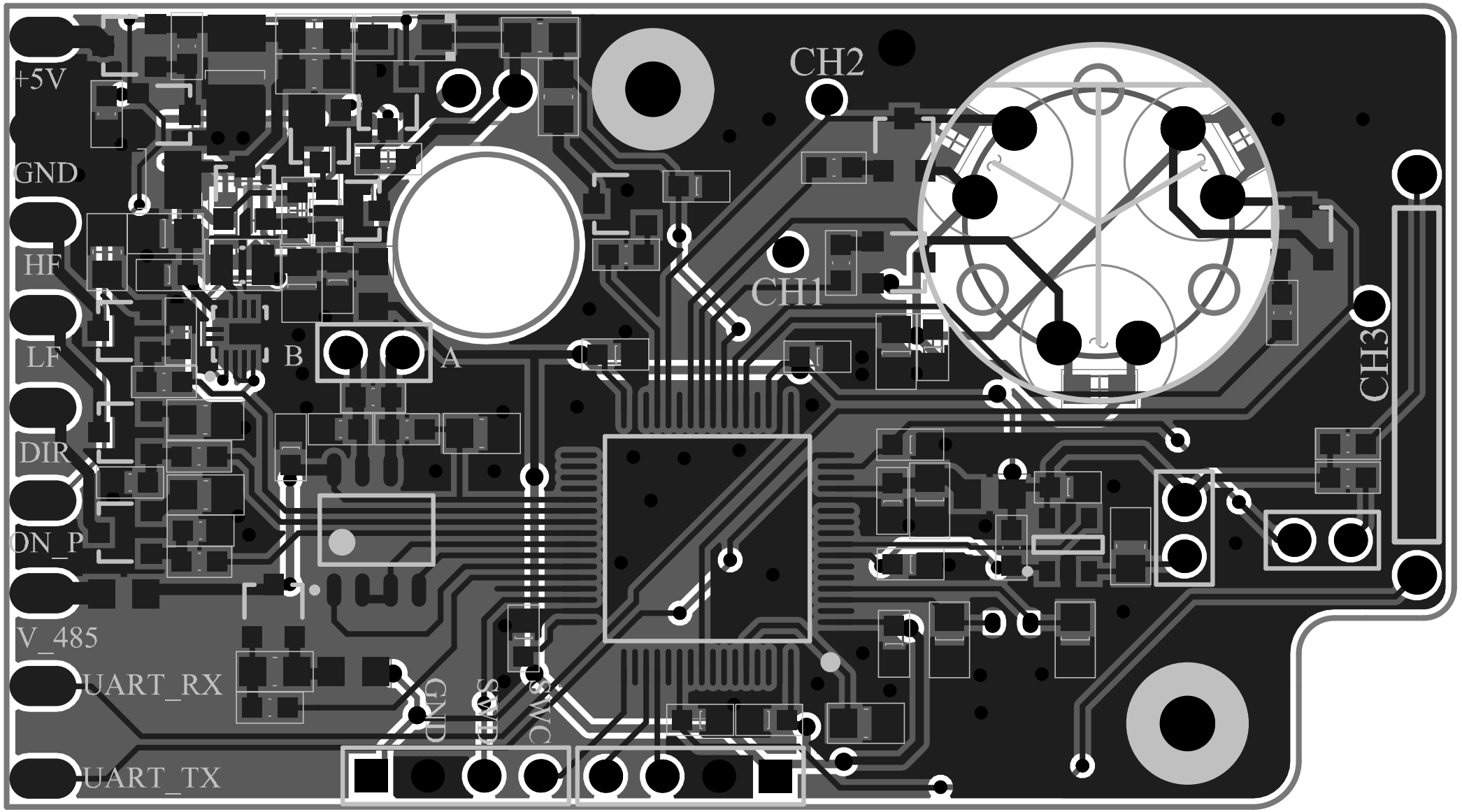


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Date: 7/05/2025		Sheet of	
File: E:\project\..\1.Power.SchDoc		Drawn By:	


U200
TWD100NYWY3F
WDO
GND
EN
VCC
WDI
NRST
Togle_Reset
R200
10k
C200
0.1uF
H200
P200
Header 4
GPS_TX
GPS_RX
C201
C202
15pF
32k_out
32k_in
Y200
R201
P201
SWC
SWD
R202
C203
C204
C205
C206
C207
C208
C209
C210
F200
VDD_A
U201
MCU
STM32L433RCT6
VBAT
PC13
PC14-OSC32_IN
PC15-OSC32_OUT
PH0-OSC_IN
PH1-OSC_OUT
NRST
PC0
PC1
PC2
PC3
VSSA/VREF-
VDDA/VREF+
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PC4
PC5
PB0
PB1
PB2
PB10
PB11
VSS
VDD
PA14
PA15
PC10
PC11
PC12
PD2
PB3
PB4
PB5
PB6
PB7
PH3_BOOT0
BOOT0
CH3
CH_3
Magnetic switch
GPS_RX
GPS_TX
SWC
VDDUSB
PA13
PA12
PA11
PA10
PA9
PA8
PC9
PC8
PC7
PC6
PB15
PB14
PB13
PB12
SWD
PW_Detect
ON/OFF_RS485
UART_RXD
UART_TXD
NET485IO
DIR_MCU
LF_MCU
HF_MCU
RS485_DT
Button
LUART_TX
LPUART_TX
LUART_RX
CH1
CH2
DAC
Clear_CH_3
V_Pin
Clear_CH_1
CH_1
Clear_CH_2
CH_2
ON_M_VPIN
3.3V
GND
VDD_A


CH_1
TP300
Clear_CH_1
R306
1k
T300
C301
102
L300
100uH
DAC
HF_OUT
HF_MCU
R303
1k
I300
R305
100k
R308
2.4R
GND
V_in_485
F300
3.3V_RS485
3.3V
F302
F301
R300
100k
ON/OFF_RS485
R302
1k
RS485_DT
R301
100k
C300
0.1uF
U300
VCC
UART_RXD
NET485IO
UART_TXD
R
RE
DE
D
GND
SN65HVD485E
R304
1k
R307
1k
R309
1k
A
B
H300
CH_2
TP301
Clear_CH_2
R313
1k
T301
C303
102
L301
100uH
LF_OUT
LF_MCU
R310
1k
I301
R312
100k
R314
2.4R
S300
R311
4.7k
Magnetic switch
C302
0.1uF
BT300
BUTTON
R315
4.7k
Button
C304
0.1uF
CH_3
TP303
Clear_CH_3
R318
1k
T302
C305
102
L302
100uH
DAC
C306
0.1uF
C307
10uF
DIR_OUT
DIR_MCU
R316
1k
I302
R317
100k
R319
2.4R
Title
Size
A4
Number
Revision
Date:
7/05/2025
File:
E:\project\..\3. Peripheral.SchDoc
Sheet of
Drawn By:


+5V
GND
HF
LF
DIR
ON_P
V_485
UART_RX
UART_TX
B
A
CH2
CH1
CH3
SAO VIET
GND
SWD
SWC
3.3V


+5V
GND
HF
LF
DIR
ON_P
V_485
UART_RX
UART_TX
B
A
CH2
CH1
CH3
GND
SWD
SWC

Comment	Manufacturer Part Number	Description	Designator	Footprint	LibRef	Quantity
0.1uF	GMC10X7R104K50NT	Cap SMT Cer .1UF 50V 10% 0603 4K/TR	C100, C102, C105, C107, C200, C203, C204, C205, C206, C207, C208, C209, C300, C302, C304, C306	C0603	C0.1uF-0805	16
1k	CR0805-8W-102JT, RC0805FR-074K7L	RES SMD 1K OHM 5% 1/8W 0805, 4.7k Ohm ±1% 0.125W, 1/8W Surface Mount Resistor Thick Film ±100ppm/°C 0805	R102, R110, R302, R303, R304, R306, R307, R309, R310, R313, R316, R318	R0603	R1k_0805, R4.7k_0805	12
1M		SMD 470K OHM 1% 1/8W 0805	R104, R106, R107, R109	R0603	R200k_0805	4
2.4R	RT0805FRE072R4L	2.4 Ohms ±1% 0.125W, 1/8W Chip Resistor 0805 (2012 Metric)	R308, R314, R319	R0805	R2.4R_0805_1%	3
4.7k	RC0805FR-074K7L	4.7k Ohm ±1% 0.125W, 1/8W Surface Mount Resistor Thick Film ±100ppm/°C 0805	R311, R315	R0603	R4.7k_0805	2
10k	RC0805JR-0710KL, RC160BJ103CS	RES SMD 10K OHM 5% 1/8W 0805, RES SMD 10K OHM 5% 1/10W 0603	R200, R201	R0805	R10k_0805, R10k_0603	2
10k	RC160BJ103CS	RES SMD 10K OHM 5% 1/10W 0603	R202	R0603	R10k_0603	1
10uF	LMK212ABJ106KG-T	10µF 10V Ceramic Capacitor X5R 0805 (2012 Metric) 0.079" L x 0.049" W (2.00mm x 1.25mm)	C101, C103, C307	C0805	C10uF_0805_10V	3
15pF	0805CG150F500NT	15pF ±1% 50V Ceramic Capacitor C0G, NP0 0805	C201, C202, C210	C0805	C15pF-0805	3
32.768k	ECS-.327-12.5-13FLX-C	32.768kHz ±10ppm Crystal 12.5pF 50 kOhm -40°C ~ 85°C Surface Mount Cylindrical Can, Radial	Y200	TF-206	Crystal_EPSON_32.768_9pF-20ppm	1
100k		RES SMD 100K OHM 5% 1/10W 0603	R101, R300, R301	R0805	R100k_0603	3
100k		RES SMD 100K OHM 5% 1/10W 0603	R305, R312, R317	R0603	R100k_0603	3
100R	RMCF0805JT100R, ERJ-U06D10R0V	RES SMD 100 OHM 5% 1/8W 0805, [NoValue]	R103, R105, R108	R0805	R100r 0805, R100R	3
100uF	GMC31X5R107M6R3NT	Ceramic Capacitor, Multilayer, Ceramic, 6.3V, 20% +Tol, 20% -Tol, X5R, 15% TC, 100uF, Surface Mount, 1206	C104, C106	C1210	C100uF_1210_10V	2
100uH	22R104C, [NoParam]	Inductor General Purpose Wirewound 100uH 10% 1KHz 90Q-Factor 670mA 390mOhm DCR RDL Bag, [NoValue]	L300, L301, L302	L_2.54_scale	L_220uH	3
102	UMK107SD102KA-T, C0603C102F5GACTU	0603 1000 pF 50 V ±10% Tolerance Standard Multilayer Ceramic Chip Capacitor	C301, C303, C305	C0805	C1nF-0805	3
200k	RT0805FRD07200KL	SMD 200K OHM 1% 1/8W 0805	R100	R0805	R200k_0805	1
700mA Step-Down	MP28301GG-P	Ultra-Low 500nA IQ, Wide Input 2-5.5V, 700mA Step-Down Regulator Plus 300nA IQ, 100mA LDO	U100	MP28301	MP28301	1
Battery Protection IC		One Cell Lithium-ion/Polymer Battery Protection IC	I100	DW01	DW01	1
BUTTON			BT300	HDR1X2	BUTTON 4 PIN	1
cong tac tu	MH002809		S300	Magnetic switch	Magnetic switch	1
Connector_9P			H100	Connector_9	Connector_9P	1
Diode Schottky 20V 1A	BAT54T1G	SCHOTTKY BARRIER DIODE Rectifier Diode, Schottky, 1 Phase, 1 Element, 0.2A, 30V V(RRM), Silicon	D100	SODFL360X90-2N	Diode Schottky 20V 1A	1
Ferrite bead 0805	CIB21P150NE	FERRITE BEAD 15 OHM 0805 1LN	F100, F200, F300, F302	L0805	Ferrite bead 0805	4
H?		hole_3mm	*100, *101	Hole_3mm	hole_4mm	2
Header 2H_j		Header, 2-Pin, Right Angle	H101	header_2P_xh2.54	Header 2H_j	1
Header 2H_j		Header, 2-Pin, Right Angle	H200, H300	HDR1X2	Header 2H_j	2
Header 4		Header, 4-Pin	P200, P201	HDR1X4	Header 4	2
IRLML6246TRPBF	IRLML6246TRPBF	Mosfet, 20V, 4.1A, 46 Mohm, 3.5 Nc Qg, 2.5V Drive Capable, SOT-23	F101, F106, F107, F108	SOT23A	IRLML6246TRPBF	4
IRML6401PbF	IRLML6401TRPBF	MOSFET P-CH 12V 4.3A SOT-23	F102, F103, F105, F301	SOT23A	IRML6401PbF	4
L_2.2uH_1.8A	CIGT201610LH2R2MNE	Inductor Power Thin Film 2.2uH 20% 1MHz Metal 1.8A 0.139Ohm DCR 0806 T/R	L100	L2016	L_2.2uH	1
Logo			Logo Sao viet100	Logo Sao viet	Logo	1
MCU	STM32L433RCT6	Ultra-low-power with FPU ARM Cortex-M4 MCU 80 MHz with 256 Kbytes Flash, LCD, USB	U201	LQFP64	STM32L433RCT6	1
MMBT3904LT1G	SMMBT3904LT1G	TRANS NPN 40V 0.2A SOT23	T300, T301, T302	SOT23A	NPN. MMBT3904LT1G	3
MOSFET N-2CH			F104	FS8052A	FS8205A	1
Mosfet N-ch 30V 10.5A		MOSFET N-CH 30V 3.4A SOT23	I300, I301, I302	SOT23A	DMG7430LFG-7	3
SN65HVD485E	SN65HVD485E	IC TXRX RS485/RS422 8-SOIC	U300	Sop 8	MAX485 SN65HVD72DR	1
TP35/55			TP300, TP301, TP303	TP_35/55	TP35/55	3
TP45/65			TP302	TP_45/65	TP45/65	1
TWD100NYWY3F	STWD100NYWY3F	IC WATCHDOG TIME CIRCUIT SOT23-5	U200	SOT23-5_L	Rst_TWD100NYWY3F	1